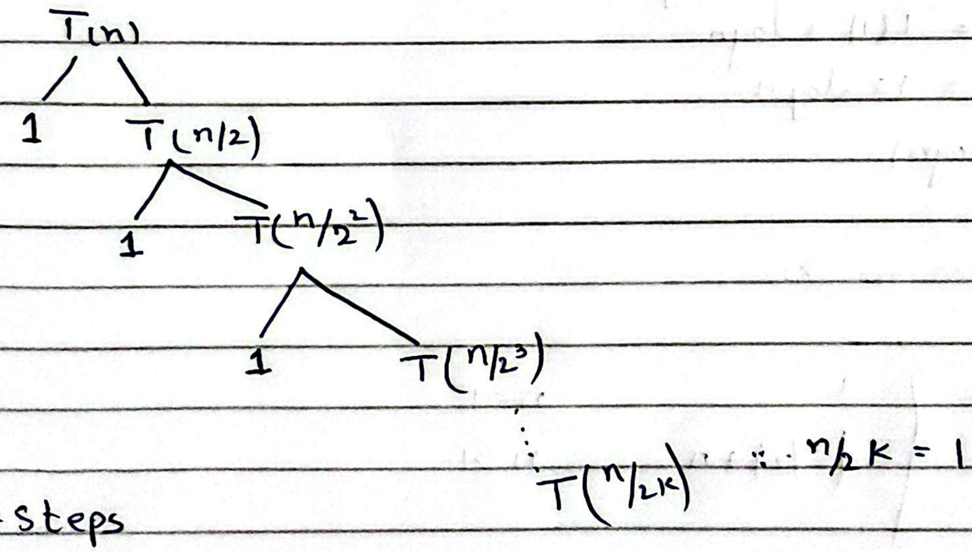
Assignment no 2
Master Theorem

Question no 1:

```
Algo Test (int n)          —— T(n)
{
    if (n>1)
    {
        Printf("%d", n);   —— 1
        Test(n/2);         —— T(n/2)
    }
}
```

$T(n) = T(n/2) + 1.$

$$T(n) = \begin{cases} 1 & n = 1 \\ T(n/2) + 1 & n > 1 \end{cases}$$



K-steps

$\therefore \frac{n}{2^k} = 1$

$n = 2^k$

$k = \log_2 n$

Now with substitution method

$$T(n) = T(n/2) + 1$$

$$T(n) = \left[T\left(n/2^2\right) + 1\right] + 1$$

$$\left[\begin{aligned} T(n) &= T(n/2) + 1 \\ T(n/2) &= T\left(n/2^2\right) + 1 \end{aligned}\right.$$

$$T(n) = T\left(n/2^2\right) + 2 \quad \text{------} \quad (2)$$

$$T(n) = T\left(n/2^3\right) + 3 \quad \text{------} \quad (3)$$

$$T(n) = T\left(n/2^k\right) + k. \quad \text{------} \quad (4)$$

Assume $n/2^k = 1$

$\therefore n = 2^k$ & $k = \log n$

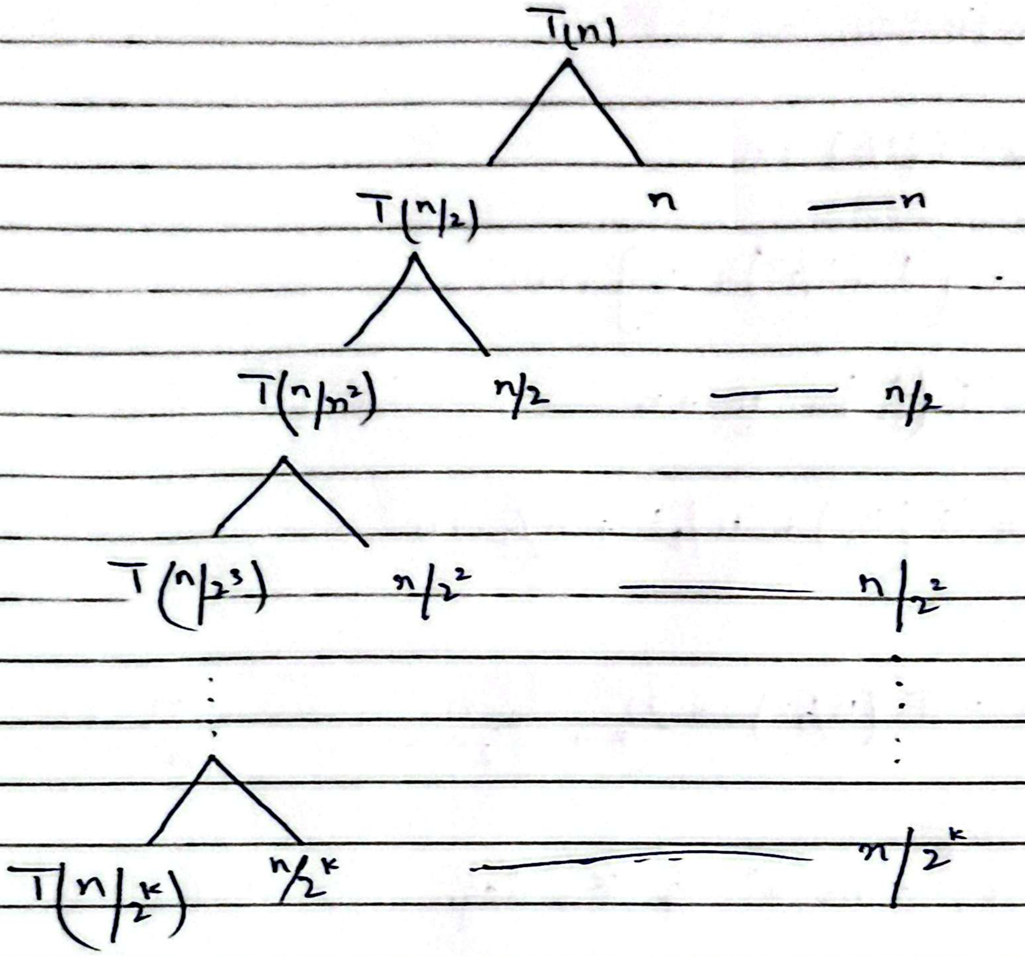
$$T(n) = T(1) + \log n$$

$$T(n) = 1 + \log n$$

$O(\log n)$

Question no 2:

$$T(n) = \begin{cases} 1 & n = 1 \\ T(n/2) + n & n > 1 \end{cases}$$



$$T(n) = n + \frac{n}{2} + \frac{n}{2^2} + \frac{n}{2^3} + \cdots + \frac{n}{2^k}$$

$$= n\left[1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^k}\right]$$

$$= n \sum_{i=0}^{k} \frac{1}{2^i} = 1$$

$$= n * 1$$

$$T(n) = n \qquad = O(n)$$

Substitution method

$$T(n) = T(n/2) + n \quad \text{--- ①}$$

$$T(n) = \left[T\left(n/n^2\right) + \frac{n}{2}\right] + n$$

$$T(n) = T\left(\frac{n}{2^2}\right) + \frac{n}{2} + n \quad \text{--- ②}$$

$$T(n) = T\left(\frac{n}{2^3}\right) + n/2^2 + n/2 + n$$

⋮

$$T(n) = T(n/2^k) + \frac{n}{2^{k-1}} + \frac{n}{2^{k-2}} + \cdots + \frac{n}{2} + n$$

Assume

$\frac{n}{2^k} = 1$, $\therefore n = 2^k$ & $k = \log n$

$$T(n) = T(1) + n\left[\frac{1}{2^{k-1}} + \frac{1}{2^{k-2}} + \cdots + \frac{1}{2} + 1\right]$$

$$T(n) = 1 + n[1+1]$$

$$T(n) = 1 + 2n$$

$O(n)$

Question no 3:

void Test (int n) —— T(n)

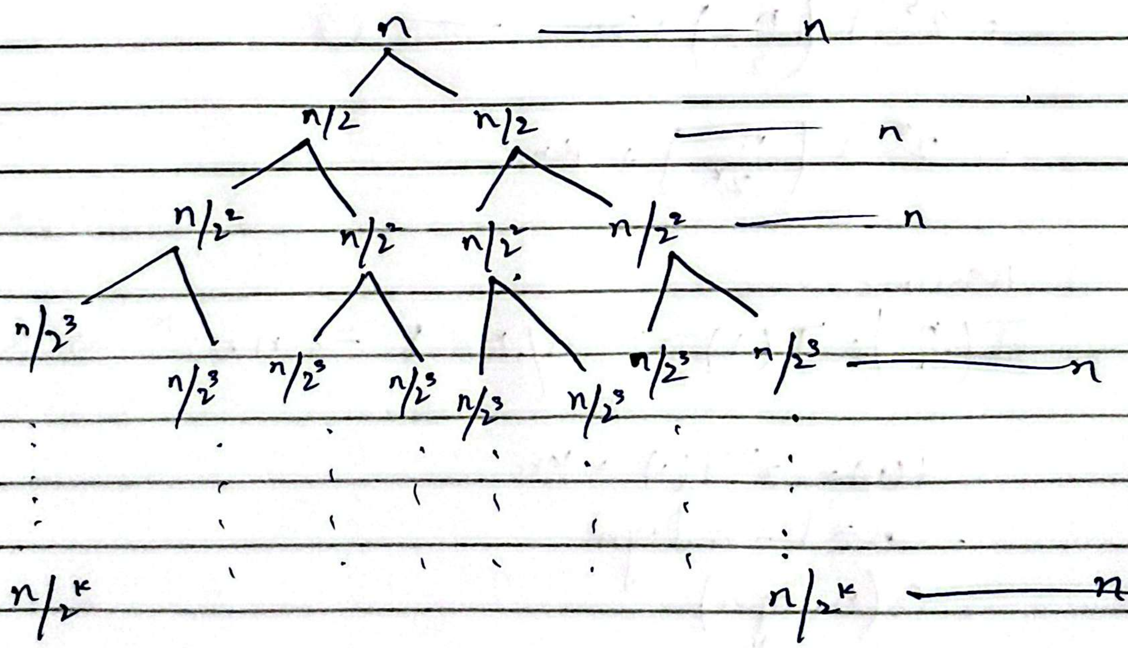
{

if (n > 1)

```
{
  for (i=0; i<n ; i++)
  {
      Stat;          ———— n
  }
  Test (n/2);        ———— T(n/2)
  Test (n/2);        ———— T(n/2)
  }
}
```

$$T(n) = 2T(n/2) + n$$

$$T(n) = \begin{cases} 1 & n = 1 \\ 2T(n/2) & n > 1 \end{cases}$$



Assume $n/2^k = 1$

$n = 2^k \qquad k = \log n$

$n \log n$

$O(n \log n)$

Also by Substitution

$T(n) = 2T(n/2) + n$ —— ①

$= 2\left[2T\left(n/2^2\right) + \frac{n}{2}\right]$

$= 2^2 T(n/2) + n + n$ —— ②

$= 2^2\left[2T\left(\frac{n}{2^3}\right) + \left(\frac{n}{2^2}\right)\right] + 2n$

$= 2^3 T\left(\frac{n}{2^3}\right) + 3n$ —— ③

$= 2^k T\left(\frac{n}{2^k}\right) + kn$

Assume

$T\left(\frac{n}{2^k}\right) = T(1) \Rightarrow n/2^k = 1 \Rightarrow n = 2^k \Rightarrow k = \log n$

$T(n) = 2^k T(1) + kn$

$= n * 1 + n \log n$

$= O(n \log n)$

Question no 4:

Cases $\rightarrow \log_b a$
$\searrow k$

~~Case 1:~~

Example 1:

$$T(n) = 2T\left(\frac{n}{2}\right) + 1$$

$a = 2$, $b = 2$

$f(n) = \Theta(1)$
$= \Theta(n^0 \log n)$

$k = 0$, $p = 0$

$\log_2 2 = 1 > k = 0$

Case 1

$\Theta(n^1)$

Example 2:

$T(n) = 4T(n/2) + n$
$\log_2 4 = 2 > k = 1$, $P = 0$
case 1:

$\Theta(n^2)$

Example 3:

$T(n) = 8T(n/2) + n$

$\log_2 8 = 3 > K = 1$, $P = 0$

Case 1

$\Theta(n^3)$

Example 4:

$T(n) = 9T(n/3) + 1$

$\log_3 9 = 2 > K = 0$ $\therefore 1 = n^0$

Case 1

$\Theta(n^2)$

Example 5:

$T(n) = 4T(n/2) + n$

$\cancel{\log_2 4}$ $\Theta(n^2)$

Example 6:

$$T(n) = 8T(n/2) + n$$

case 1

$$\Theta(n^3)$$

Example 7:

$$T(n) = 8T(n/2) + n \log n$$

case 1

$$\Theta(n^3)$$

Case 2:

Example 1:

$$T(n) = 2T(n/2) + n$$

$\log_2 2 = 1$ $\quad$ $k = 1$, $p = 0$

$$\Theta(n \log n)$$

Example 2:

$$T(n) = 4T(n/2) + n^2$$

$\log_2 4 = 2 \Rightarrow k = 2$

$$\Theta(n^2 \log n)$$

Example 3:

$$T(n) = 4T(n/2) + n^2 \log n$$

$\log_2 4 = 2 = k = 2$

$$\Theta(n^2 \log^2 n)$$

Example 4

$$T(n) = 4T(n/2) + n^2 \log^2 n$$

$\log_2 4 = 2 = k = 2$

$$\Theta(n^2 \log^3 n)$$

Example 5

$$T(n) = 4T(n/2) + n^2 \log^5 n$$

$\log_2 4 = 2 \quad = \quad k = 2$

$$\Theta(n^2 \log^6 n)$$

Example 6

$$T(n) = 8T(n/2) + n^3$$

$\log_2 8 = 3 \qquad k = 3$

$$\Theta(n^3 \log n)$$

Example 7

$T(n) = 2T(n/2) + n/\log n \qquad \left(\frac{n}{\log n} = n \log^{-1} n.\right.$

$\log_2 2 = 1 \qquad k = 1 \qquad P = -1$

$$\Theta(n \log \log n)$$

Example 8

$$T(n) = 2T(n/2) + n \log^{-2} n$$

$\log_2^2 = 1 \quad k = 1 \quad p = -2$

$\Theta(n) \quad \because \text{ if } p < -1 \Rightarrow \Theta(n^k)$

Example 9

$T(n) = T(n/2) + n^2$

$\log_2^1 = 0 < k = 2$

$\Theta(n^2)$

Example 10:

$T(n) = 2T(n/2) + n^2$

$\log_2^2 = 1 < k = 2$

$\Theta(n^2)$

Example 11:

$T(n) = 2T(n/2) + n^2 \log n$

$\log_2^2 = 1 < k : 2$

$\Theta(n^2 \log^2 n)$

Example 13:

$T(n) = 4T(n/2) + n^3$

$\log_2^4 = 2 < K = 3$

$\Theta(n^3)$

Example 14

$T(n) = 4T(n/2) + n^3 / \log n)$

$\log_2^4 = 2 \quad < K = 3$

$\Theta(n^3)$

Question no 5:

Case 1 :-

Q 1:

$T(n) = 2T(n/2) + 1 \quad - \Theta(n)$

Q2:

$$T(n) = 4T(n/2) + 1$$

$$\Theta(n^2)$$

Q3:

$$T(n) = 4T(n/2) + n$$

$$\Theta(n^2)$$

Q4:

$$T(n) = 8T(n/2) + n^2$$

$$\Theta(n^3)$$

Q5:

$$T(n) = 16T(n/2) + n^2$$

$$\Theta(n^4)$$

case2 Q1:

$$T(n) = T(n/2) + 1$$

$$\Theta \log n$$

Q2:

$$T(n) = 2T(n/2) + n$$

$$\Theta(n \log n)$$

Q3:

$$T(n) = 2T(n/2)\ n \log n$$

$$\Theta(n \log^2 n)$$

Q4:

$$T(n) = 4T(n/2) + n^2$$

$$\Theta(n^2 \log n)$$

Q5:

$$T(n) = 4T(n/2) + n \log n)^2$$

$$\Theta(n^2 \log^3 n)$$

Q6:

$$T(n) = 2T(n/2) + n^1/\log n$$

$$\Theta(n \log \log n)$$

Case 3:

Q1:

$$T(n) = T(n/2) + n$$

$$\Theta(n)$$

Q 2:

$$T(n) = 2T(n/2) + n^2$$

$$\Theta(n^2)$$

Q 3:

$$T(n) = 2T(n/2) + n^2 \log n$$

$$\Theta(n^2 \log n)$$

Q4:

$$T(n) = 4T(n/2) + n^3 \log^2 n$$

$$\Theta(n^3 \log^2 n)$$

Q5:

$$T(n) = 2T(n/2) + n^2/\log n$$

$$\Theta(n^2)$$